

with pre-loaded reagents, where disease-specific zones provide an objective read-out, much like an ELISA assay.

A small device that easily connects to a smartphone or computer was recently piloted by healthcare workers in Rwanda, who tested blood obtained via a finger prick from 96 patients.

During the field testing in Rwanda, the vast majority of patients (97 per cent) said that they would recommend the device because of its fast turn-around time, ability to offer results for multiple diseases and simplicity of procedure.

This touchscreen touches you back

Touching your smartphone or tablet is such a *passé*! Yes, it is. It is time for touchscreens to touch you back and yes, we are not talking of Hollywood fiction here.

As per researchers from Northwestern University, the USA, and Carnegie Mellon University, the USA, any flat surface has two virtual bumps, but under certain circumstances, people touching the surface feel only one bump. Fingers might get an illusion that the surface actually contains bumps, due to the forces they feel. This is termed as virtual bump illusion, and is a well-known term in the touch based (haptics) field.



The touchscreen of your smartphone or tablet could touch you back

Explaining about the research, J. Edward Colgate, an expert in haptic systems from Northwestern's McCormick School of Engineering and Applied Science, stated that they aim to create something that will make touch a reality for people interacting with their screens, and this work is a step in that direction.

The team presented two virtual bumps to people participating in the study. Subjects of the study reported two bumps as one when the spacing between bumps and finger were identical. In such a case, the brain tends to think of a coincidental situation where there should be two bumps at the same time so it registers the bumps as one.

This kind of experiment can help in creating flat-screen displays with active touch-back technology, which will make a touchscreen keyboard feel like an actual keyboard.

A fire-fighting robot to help the Navy

Students from Virginia Institute of Technology, the USA, have built a human-like robot called SAFFIR. The robot is specially designed to help sailors of the Navy in their missions. The team has been working on the prototype of SAFFIR (shipboard autonomous fire-fighting robot) for four years. SAFFIR is capable of helping sailors in hazards like fire.

The prototype has undergone various tests to ensure reliability. SAFFIR can handle all major fire problems very efficiently. It weighs 64kg and is the size of a human being. It has two legs and two hands, and can pick up and handle all fire-fighting instruments. The robot can operate a water hose, which can be used to spray water in case of fire.

In hazardous situations, human beings have their limitations. Robots like SAFFIR can be amazing non-human entity managers to carry out human-like tasks. The robot's body is made of a strong shell, which can withstand flames and the heat of a fire. It can accurately judge victims of fire and rescue them. SAFFIR is equipped with infra-red eyes that let it see even through the smoke. The robot is designed with light detection and ranging (LIDAR) technology.

The team is working on an advanced version of this robot that will feature surplus memory and faster sensor equipment.

A robot scientist that can speed up drug recovery process

Automation is entering the field of science with great speed. Robotic scientists are now capable of conducting experiments, study observations, analyse and explain findings, and they never fall tired till the results look satisfactory enough. Hence, hypothesis-led automated research work has received a huge boost.

A team of researchers from University of Manchester has developed a robot scientist, EVE, which can speed up the process of drug recovery. EVE can identify new drug candidates for deadly diseases like malaria and others, including African sleeping sickness and Chagas' disease.

These diseases infect millions of people worldwide. The cost and speed required to discover small molecule drugs for these diseases are almost impossible. Now, with the help of EVE, researchers can use artificial intelligence technology to select, screen and separate compounds, which can discover drugs in a faster way. EVE can automate the early-stage drug-designing process, and its robotic system can also screen over 10,000 compounds every day.

EVE can also improve the screening process by selecting a sub-set of the library to find compounds. Researchers believe that a future version of EVE would be able to synthesise such compounds and not just discover these.

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